Madugula Murthy Manideep

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.73
Sri Chaitanya Junior College, Vijayawada	Class XII (Board of Intermediate Education Andhra Pradesh, India)	2024	97.60%
Sri Chaitanya High School, Vijayawada	Class X (Board of Secondary Education Andhra Pradesh, India)	2022	96.33%

SCHOLASTIC ACHIEVEMENTS

- Received the LPUNEST Merit Scholarship for securing a good rank in the LPUNEST examination.
- Received the SBI Asha Scholarship (2025) based on academic merit.

SKILLS

- **Languages:** C, C++, Java, JavaScript, Assembly(Risc-V), SQL
- **Development:** React, Express
- **Operating Systems:** Ubuntu
- **Databases:** MySQL, PosgreSQL
- **Dev Tools:** Docker, Git, Github, Postman
- **Others:** LaTex, Vim, SolidEdge

RELEVANT COURSES

- **Computer Science:** Introduction to Programming, Discrete Mathematics, Software Development Fundamentals, Operating Systems, Computer Architecture, Database Management Systems, Data Structures, Algorithms, Theory of Computation, Introduction to Compilers
- **Mathematics :** Calculus, Linear Algebra, Probability and Random Variables, Transform Techniques, Introduction to Statistics.
- **Others:** Introduction to Artificial Intelligence, Digital Fabrication.

PROJECTS

OS Process Scheduling and Memory Management | GitHub Link

- **Implemented custom system calls** and a **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler** with **dynamic priority**, **time-slice scheduling**, and **periodic priority boosting** to prevent starvation.
- Implemented an **in-memory swap mechanism** with **lazy page allocation**, **page fault handling**, **Clock-based page replacement**, **scheduler-aware page eviction**, and **page swap-in/swap-out operations**.
- **Added RAID-backed swap storage** to replace the in-memory swap system, implementing **fault-tolerant swap management** through **RAID-based read/write operations** for reliable virtual memory.

Reader-Writer Synchronization using Semaphores | GitHub Link

- Implemented **Fair Reader-Writer** and **Writer Preference** synchronization using **POSIX semaphores** and **threads**.
- Compared their performance by measuring the **average** and **worst-case waiting time** to enter the **Critical Section**.

Computer Architecture Optimization using FPU Registers | Drive Link

- Implemented **5×5 blocked matrix multiplication** in **RISC-V assembly** using **loop unrolling**, computing **25 output elements simultaneously** while supporting **arbitrary matrix sizes** through boundary checks.
- Replaced **stack-based accumulators** with **25 FPU registers** using only **fmv instructions**, eliminating repeated **ld/sd operations** while preserving identical integer computation.
- Evaluated both implementations on **gem5** under different **cache latencies**, demonstrating reduced **memory accesses**, fewer **memory stall cycles**, and improved performance as **cache latency** increased.

Arbitrary Precision Arithmetic Library | Github Link

- Implemented **arbitrary-precision integer and floating-point arithmetic** using **string-based representation** with custom **addition, subtraction, multiplication, and long division** algorithms.
- Supported **arbitrary-length numbers**, **30 decimal places of precision**, **sign handling**, **decimal alignment**.
- Packaged the project as a reusable **Java library (JAR)** and **Docker** support for easy to use into other Java applications.

WhiteBoard Application | GitHub Link

- Built an **interactive canvas-based whiteboard** using **React, Redux Toolkit, Rough.js, and Canvas**, implementing **shape drawing, text editingcopy/paste, undo/redo history, localStorage and PNG export**.
- Integrated **Gemini AI sketch-to-image generation** through an **Express.js backend**, securely handling API keys while allowing users to transform hand-drawn sketches into realistic images and continue editing them on the canvas.

NSS Website | GitHub Link

- Built a basic NSS website using React as part of the Web Coordinator selection task.
- Developed an Express.js backend and PostgreSQL database to store and manage student NSS hours.

POSITIONS OF RESPONSIBILITY

- Web Coordinator at National Service Scheme (NSS), IIT Hyderabad